MIL-H-17656E(NU)
6 February 1984
SUPERSEDING
MIL-H-17656D(SA)
22 November 1977

MILITARY SPECIFICATION

HAT, SOU'WESTER

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a waterproof foul weather hat made of a rubber coated nylon twill cloth.

1.2 Classification. The hat shall be of one type only and in the following sizes (see 6.2):

Small	- Up to 6 3/4
Medium	- 6 7/8 to 7
Large	- 7 1/8 to 7 1/4
Extra Large	- 7 3/8 and over

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the soliticiation form a part of this specification to the extent specifed.

SPECIFICATIONS

FEDERAL

V-T-295	Thread, Nylon
DDD-L-20	Label; for Clothing, Equipage, and Tentage (General Use)
DDD-T-86	Tape, Textile, Cotton, General Purpose (Unbleached, bleached or dyed)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8405

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MIL-B-17757	Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-C-19759	Cloth, Coated, Twill, Nylon, (Low Count)
MIL-C-43718	Cloth, Twill, Polyester and Cotton; Polyester and Rayon

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publication. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Ill 60606.)

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3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified, (see 6.2) the Contractor shall furnish sample unit(s) for the first article inspection and approval (see 4.2).

3.3 Materials

3.3.1 Basic material. The basic material for the hat shall be a chloroprene coated nylon twill fabric conforming to Type II of MIL-C-19759, except that the coating shall be unvulcanized.

* 3.3.2 Cloth, corduroy, cotton. The material for the neck shield shall be 100% cotton or a cotton/polyester blend of not less than 80% cotton. It shall have a minimum of eight wales per inch and weigh not less than 8 oz. per sq. yd. The cloth shall be fast dyed (no sulphur) dark green, black or olive drab and after treated with a durable water repellent impregnant (see 6.4). The dyed and treated cloth shall show good colorfastness to crocking and perspiration.

3.3.3 Cloth, twill, polyester. The twill cloth used for lining the crown shall be dyed black or olive green and shall conform to type I of MIL-C-43718.

3.3.4 Tape, cotton. The cotton tape used for the tie tape shall be 3/4 inch wide conforming to type I, class 3, of DDD-T-86. The color shall be black or olive green.

3.3.5 Cement, chloroprene. The cementing compound shall have a chloroprene base and the same curing characteristics as the coating compound specified in 3.3.1. The use of reclaimed rubber is prohibited. Cemented sample seams shall have a minimum adhesion of 5 pounds per inch width after vulcanizing when tested as specified in 4.4.1d. The color of the cement shall approximate the shade of the green coating compound specified in 3.3.1 or shall be unpigmented.

* 3.3.6 Label. Each hat shall have a combination identification-size label conforming to type VI, class 4, of DDD-L-20; except that the label shall bear the following inscription:

HAT, SOU'WESTER
CONTRACT NO: DLA-100-00-0-000 (Example)
STOCK NO: 8405-00-000-0000 (Example)
SIZE: MEDIUM (6 7/8 to 7) (Example)
NAME OF CONTRACTOR:

3.3.7 Thread, nylon. The nylon thread shall conform to type III, size B, or E of V-T-295; except that the colorfastness requirements shall not apply. The color shall be olive drab S-1, C.A. 66022 or shall approximate the green shade of the basic material.

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3.3.8 Dusting powder. The dusting powder used in coating the basic material and for fabricating the hat shall be a suitable dusting medium that does not support mildew growth in the finished vulcanized hat (see 3.6.2 and 4.4.1.1).

3.4 Design. The hat shall be of sou'wester design; with the crown, brim, and outer neck shield made of a coated nylon fabric. A corduroy lined fold down neck shield shall be positioned on the inside at back of hat for neck and ear protection. Tie tapes shall be attached to the neck shield for securing hat underneath the chin. The crown of the hat shall be lined with twill cloth.

3.4.1 Figure. Figure 1 is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns will be furnished by the Government to the contractor for use in cutting working patterns (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

3.5.1 Pattern parts. The component parts shall be cut from the material specified in accordance with the following pattern parts:

Materials	Nomenclature of pattern parts	Cut parts
Cloth, coated, Nylon twill	Top crown	1
	Side crown	1
	Brim (upper and lower)	2
	Neck shield	1
Cloth, twill, polyester, polyester and cotton, polyester and rayon	Top crown lining	1
	Side crown lining	1
Cloth, corduroy, cotton	Neck shield	1
Template for cementing	Brim (outer edge)	

3.6 Construction

3.6.1 Sewn seams. Stitch and seam types specified in table I shall conform to FED-STD-751. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats, or open seams will result. Ends of all seams and stitchings shall be backtacked not less than 1/2 inch. Thread breaks shall be secured by starting the repair stitch not less than 1/2 inch in back of the break. The minimum and maximum number of stitches per inch shall be as specified in table I.

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3.6.2 Cemented seams and parts. Prior to the joining of precemented and dusted parts the surfaces to be joined shall be washed with a suitable solvent and allowed to dry fully. Care shall be exercised in performing this operation to avoid solvent staining of adjacent surfaces. The width of the visible cement line for seams after joining shall be a maximum of 3/8 inch on the outside and 3/8 inch on the underside with no feathering or loose edges. Wherever possible, cementing shall be performed with the aid of templates to insure that cement lines are even, uniform, and within specified tolerances. Where cemented seam types are specified in table I, they are similar to the seam types indicated in FED-STD-751, minus the stitchline. All seams shall be rolled to remove any puckers, tunnels or wrinkles. Before vulcanization, the seams shall be dusted with the dusting powder specified in 3.3.8, using the minimum amount necessary to prevent adhesion to other surfaces.

3.7 Manufacturing operations requirements. The hats shall be manufactured in accordance with operation requirements specified in table I. The contractor is not required to follow the exact sequence of operations shown.

3.7.1 Shade and size marking. The component parts shall be marked, ticketed or bundled to insure a uniform shade and size throughout the hat. Any method may be used except:

1. Metal fastening devices.
2. Sew-on tickets.
3. Adhesive type tickets whose adhesive causes discoloration or whose adhesive mass adheres to the material upon removal of the tickets.

3.7.2 Vulcanization. The hats shall be vulcanized in accordance with the time and temperature requirements supplied by the coated fabric manufacturer.

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Table I - Construction of Hat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting a. Cut the hats in strict accordance with patterns furnished which show size, shape, directional lines for cutting, and notches for proper assembly of all parts. b. All basic fabric component parts shall be cut from one piece of material. The lightly coated (green) side of the basic material shall finish on the outside of the hat. c. The use of a drill hole is permitted only when the hole is located in an area where it will be covered in a later operation by the cement portion of a cemented component part. d. All other component parts shall be cut in accordance with patterns or in accordance with description of operation when no pattern is furnished.					
2.	Replacement of defective components During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4 of this specification, shall be removed from production and replaced with non-defective and properly matched components.					
3.	Shade and size marking (see 3.7.1) a. All parts shall be marked, ticketed or bundled to insure uniformly shaded and properly sized hats throughout all assembly operations.					

NCTR Form 1, 15 Sep 1967

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Table I - Construction of Hat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make brim					
	a. Turn under outer edge of upper brim 1/4 to 3/8 inch and stitch 1/8 to 3/16 inch from folded edge (See NOTE:).	301	EFa-1	8-10	B or E	B or E
	b. Apply a 3/4 inch cement line to the front seam (each side) of upper brim and join ends by overlapping left side on right side 5/8 to 3/4 inch		LSa			
	c. Turn under outer edge of lower brim 1/4 to 3/8 inch and stitch 1/8 to 3/16 inch from folded edge edge (See NOTE:).		EFa-1	8-10	B or E	B or E
	d. Apply a 3/4 inch cement line to the front seam (each side) of lower brim and join ends by overlapping right side on left side 5/8 to 3/4 inch.		LSa			
	e. Apply cement to outer edges of upper and lower brims on back side, using template, and join together in accordance with marks on pattern.		SSe (b)			
	NOTE: As an alternate, the outer edge may be turned under and cemented.		EFa			
5.	Make crown lining					
	Stitch face to face, the top crown lining piece to the side crown lining piece 1/4 to 3/8 inch from the raw edge.	301	SSa-1	8-10	B or E	B or E

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Table I - Construction of Hat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
6.	Attach label Center the label on the top crown lining with the top edge of label approximately 2 inches from the front seam of lining and stitch all four sides.	301	LSbj-1	8-10	B or E	B or E
7.	Make neck shield					
	a. Stitch darts in corduroy neck shield piece in accordance with marks on pattern.	301	LSq-2(a)	8-10	B or E	B or E
	b. Fold back seam allowances towards top of neck shield and topstitch dart 1/16 to 1/8 inch from folded edge.	301	LSq-2(b)	8-10	B or E	B or E
	c. Stitch darts in coated fabric neck shield piece in accordance with marks on pattern.	301	LSq-2(a)	8-10	B or E	B or E
	d. Fold back seam allowances towards top of neck shield and topstitch dart 1/16 to 1/8 inch from folded edge.	301	LSq-2(b)	8-10	B or E	B or E
	e. Stitch neck shield pieces (corduroy and coated fabric) together, face to face, along sides and bottom 1/4 to 3/8 inch from raw edge, catching width of tie tapes in the seam as indicated by marks on patterns. Turn and edgestitch 1/16 to 1/8 inch from folded edge.	301	SSe-2	8-10	B or E	B or E
	f. The finished free length of the tie tapes shall measure 10 1/2 to 11 1/2 inches in length, with the ends cut on the bias to prevent fraying.					

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Table I - Construction of Hat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Join assemblies					
	a. Position, as worn, the coated side of the neck shield assembly against the underside of the brim assembly in the center back area, as indicated by marks on pattern, and stitch 3/16 to 5/16 inch from raw edge.	301	SSa-1	8-10	B or E	B or E
	b. Superimpose, face to face, the raw edge of crown lining assembly and combined brim and neck shield assemblies at center back notch, as indicated on patterns, and stitch entire circumference of hat opening 1/4 to 3/8 inch from raw edge.		SSa-1	8-10	B or E	B or E
9.	Make crown outershell					
	Apply a 1/2 inch cement line to top and side crown pieces and join together by lapping the top piece over the side piece 1/4 to 3/8 inch.		LSa			
10.	Join crown to brim					
	Apply a 5/8 inch cement line to the bottom edge of side crown piece and inner edge of upper brim and join together by lapping side crown over the upper brim 3/8 to 1/2 inch.		LSa			
11.	Vulcanization					
	The assembled hat shall be fully vulcanized using heated dry air rooms or pressure vulcanizers. (see 3.7.2)					

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3.8 Sizes and measurements. Sizes and measurements of the finished hats shall be as shown in table II. All measurements and tolerances are expressed in inches.

Table II - Sizes and finished measurements

	A	B	C	D	E	F	G
Sizes	Brim Front Center	Brim Back Center	Height Crown (Front)	Length Neck Shield	Width Neck Shield (Front)	Width Neck Shield (Back)	Brim Length
Small 6 3/4	2 3/4	5 1/4	3 1/4	13 7/8	4 1/4	3 3/4	33 3/4
Medium 6 7/8 to 7	2 3/4	5 1/4	3 1/4	14 1/4	4 1/4	3 3/4	34 1/2
Large 7 1/8 to 7 1/4	2 3/4	5 1/4	3 1/4	14 5/8	4 1/4	3 3/4	35 1/4
X-Large 7 3/8	2 3/4	5 1/4	3 1/4	15	4 1/4	3 3/4	36
Tolerance	± 1/4	± 1/4	± 1/4	± 1/4	± 1/4	± 1/4	± 3/8

NOTE: Refer to figure 1 for A, B, C, D, E, F, and G measurements.

A. Measurement taken along front center seam from base of crown to outer edge of brim.

B. Measurement taken from base of crown to outer edge of brim at center back.

C. Measurement taken at center front seam from base of crown to top of crown.

D. Twice the measurement taken along center of shield, with shield folded in half, from front edge to back edge. The darts shall indicate the center.

E. Measurement taken with shield turned down from seamline at top of shield to bottom folded edge of shield at inner end of dart.

F. Measurement taken with shield turned down, from seamline at top of shield to bottom folded edge of shield at center back.

G. Twice the measurement taken along length of brim at the widest point with brim folded in half.

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* 3.9 End item performance testing. The vulcanized cemented seams shall conform to the requirements listed in table III when tested as specified in 4.5. The requirements listed do not refer to stitched seams.

Table III - Requirements for cemented seams in end item

Characteristics	Requirements Average (min)	Requirements Individual (min)
Seam Strength - lbs	150	100
Seam Adhesion - lbs/inch width	5.0	4.0
Hydrostatic resistance - lbs/sq in		
Initial	100	50
After 3 launderings	80	50
After 1 dry cleaning	80	50

3.10 Coating repairs on hat

3.10.1 Black coated side (inside). Pinholes and uncoated areas of any type generated during the manufacture of the hat shall be repaired either before or after vulcanizing the assembled hat with the black coating compound specified in 3.3.1. When hats are repaired after curing, the repair compound shall be formulated to cure at room temperature. The size of the inside repair shall not exceed 3/4 inch in length or width and the number of repairs per hat shall not exceed two. The repairable area shall be coated with a minimum of two coats of the repair compound. The first coat shall completely wet out the defective area and shall be thoroughly worked into the surface. The second coat shall be spread on smoothly and evenly over the first coating. After the compound is dry the repaired area shall be lightly dusted with the dusting powder specified in 3.3.8. When fully cured there shall be no lifting of the edges, cracking, flaking or removal of the repair compound when the inside repaired area is subjected to a flexing action between the hands.

3.10.2 Green coated side (outside). Any area where the nylon base fabric is exposed shall be touched-up (hidden), before or after curing, with a suitably thinned solution of the green coating compound specified in 3.3.1. When hats are repaired after curing, the touch-up solution shall be formulated to cure at room temperature. Only the minimum amount of solution necessary to hide the uncoated area shall be applied. The number of repairs allowed per hat on the outer green coated side shall not exceed two. The color of the repaired areas shall be a good match to the surrounding coating.

3.11 Workmanship. The finished hat shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article, submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, dimensional requirements, and end item testing.

4.4 Inspection. Inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of components. In addition, components listed in table IV shall be tested for the characteristics specified and in accordance with the referenced test methods of FED-STD-191, unless otherwise specified. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The unit of product for lot formation and testing shall be as shown below:

COMPONENT	LOT SIZE EXPRESSED AS	SAMPLE UNIT FOR TESTING
Cloth, Coated, Nylon Twill (Vulcanized)	yards	3 continuous yards, full width

The sample size shall be in accordance with the following:

Lot Size	Sample Size
800 or less	2
801 through 22,000	3
22,001 and over	5

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Table IV - Testing of components

Component	Characteristic	Requirement paragraph	Test Method
Cloth, Coated, Nylon Twill (Vulcanized)	Blocking	3.3.1	1/
	Breaking strength	3.3.1	1/
	Weight	3.3.1	1/
	Tear strength	3.3.1	1/
	Hydrostatic resistance	3.3.1	1/
	Flex stiffness	3.3.1	1/
	Adhesion of coating	3.3.1	1/
	Resistance to heat	3.3.1	1/
*Cloth, Corduroy	Material Identification	3.3.2	2/
	No wales	3.3.2	2/
	Weight	3.3.2	2/
	Color	3.3.2	2/
	Water repellent treatment	3.3.2	2/
Cement, Chloroprene	Identification	3.3.5	3/
	Seam adhesion	3.3.5	5960 3/
Dusting powder	Resistance to mildew growth	3.3.8	2/

1/ Three yards of unvulcanized cloth shall be vulcanized under conditions similar to those used on the end item and shall be tested to meet the requirements specified in paragraph 3.3.3.4 of MIL-C-19759 for type III vulcanized cloth.

2/ A certificate of compliance shall be submitted for this requirement.

3/ Cemented seam specimens shall be prepared in accordance with 3.6.2, except that the seam shall overlap 1 1/4 inches. After vulcanization a 1 inch wide by 6 inch long strip shall be marked and cut from the cemented seam in the length direction.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one finished hat. The lot size shall be expressed in units of hats.

4.4.2.1 Visual examination. The hats shall be examined for defects classified in accordance with the list below:

Defect	Classification Major	Minor
I. MATERIAL DEFECT(S) AND DAMAGE(S)		
A. Corduroy Cloth		
1. Any hole, needle chew, cut, tear or other type weakening defect	X	
2. Any visible mend or burn	X	

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Defect	Classification Major	Minor
B. Coated Cloth		
1. Any stitched mend or cemented cut patch	X	
2. Any hole, cut, tear or burn	X	
3. Any pinhole, blister, lump or imbedded foreign matter	X	
4. Any weak or thin area	X	
5. Abraded or scuffed, resulting in a weak place or coating removal	X	
6. Permanent crease or wrinkle	X	
7. Any pick off or uncoated area	X	
8. Any tackiness	X	
9. Coating soft, not fully vulcanized	X	
10. Any brittle area coating cracks when bent	X	
11. Any objectionable odor	X	
II. CLEANNESS		
1. Any permanent spot or stain on outside		X
2. Two or more spots stains or shade, size markings not removed on outside		X
3. Thread ends not trimmed to less than 1/2 inch, three or more		X
4. Cement on any area other than where specified	X	
5. Excessive dusting powder distributed over seams, outside only		X
6. Excessive solvent staining on outside		X
III. COMPONENT AND ASSEMBLY (Unless otherwise classified)		
1. Any component omitted, misplaced, damaged or otherwise defective	X	
2. Any required operation omitted	X	
3. Any required operation not as specified		X
4. Any component other than type specified		X
IV. SHADED PART(S)		
Any outside part badly shaded		X
V. CUTTING		
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements.	X	
VI. BRIM		
1. Center front seam of upper or lower brim less than 5/8 inch or more than 3/4 inch		X

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Defect	Classification Major	Minor
2. Outer edges of upper or lower brim turned under less than 1/4 inch or more than 3/8 inch		X
VII. CROWN OUTERSHELL		
1. Top crown piece overlapping side crown piece by less than 1/4 inch or more than 3/8 inch		X
2. Bottom of side crown piece overlapping upper brim by less than 3/8 inch or more than 1/2 inch		X
VIII. IDENTIFICATION LABEL		
1. Omitted, incorrect or illegible	X	
2. Misplaced		X
3. Not securely attached		X
IX. TIE TAPES		
1. Not type specified		X
2. Width of tape not as specified		X
3. Any omitted	X	
4. Finished free length less than 10 1/2 inches or more than 11 1/2 inches		X
5. Ends of tape not bias-cut		X
6. Not securely caught in seam		X
X. SEAMS AND STITCHING		
A. General		
1. Any cemented or stitched seam twisted, puckered, or pleated	X	
2. Any part of hat caught in an unrelated cementing or stitching operation.		X
B. Stitched seams		
1. Stitching margin not as specified		X
2. Ends not securely tacked or not tacked as specified		X
3. Any open seam (see Note:)		
(a) 1/4 to 1/2 inch, inclusive		X
(b) More than 1/2 inch	X	
4. Loose tension resulting in a loosely secured seam		X

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Defect	Classification Major	Minor
5. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X	
Note: One or more broken stitches or two or more continuous skipped or runoff stitches constitute an open seam.		
C. Stitches per inch (score only when condition exists on more than 1/2 the length of seam)		
1. One stitch less than the minimum or any number in excess of the maximum specified		X
2. Two or more stitches less than the minimum specified.	X	
D. Seam and stitch types		
1. Not specified seam or stitch type	X	
2. Any line of stitching not beginning or not ending where specified		X
3. Any required line of stitching omitted	X	
E. Cemented seams		
1. Poorly rolled forming:		
a. An open tunnel		
(1) extending for up to 1/2 the width of the seam		X
(2) extending for more than 1/2 the width of the seam	X	
b. A permanent pucker, wrinkle, fold or closed tunnel		X
2. Direction of seam lap other than specified		X
3. Any loose or unaffixed edge with a:		
a. Length		
(1) up to 1/4 inch inclusive		X
(2) more than 1/4 inch	X	
b. Depth		
(1) up to 1/8 inch inclusive		X
(2) more than 1/8 inch	X	
4. Visible cement line more than 3/8 inch wide for more than 1/2 the length of seam		X
5. Excessive cement, resulting in adhesion of surfaces that were not required to be cemented	X	
6. Dusting powder omitted or improperly applied causing adhesion between two surfaces	X	

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Defect	Classification Major	Minor
XI. REPAIRED AREA(S)		
1. Color of repaired area on green side not good color match		X
2. Repair compound on coated fabric cracks, lifts off or flakes when coated fabric and repaired area are flexed between the hands		X
3. Repair compound not smoothly or evenly applied		X
4. Repair compound does not cover defective area		X
5. Repaired area not dusted		X
6. Size of repair area more than 3/4 inch in length or width		X
7. Repaired areas number more than two on the inside or more than two on the outside		X

4.4.2.2 Dimensional examination. The finished hat shall be examined for defects in dimensions as specified in table II. Any measurement deviating from nominal dimensions and tolerances specified shall be scored as a measurement defect.

4.4.2.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection Level	AQL's Major	AQL's Total
For visual examination in 4.4.2.1	II	2.5	6.5
For dimensional examination in 4.4.2.2	S-3	-	4.0

4.4.3 Packaging examination. An examination shall be made to determine that the packaging, packing and marking complies with Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

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Examine	Defect
Marking (exterior or interior)	Omitted, incorrect, illegible, of improper location, sequence, or method of application.
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling, improper taping. Bulged or distorted container.
Content	Number of hats per shipping container is more or less than required. 1/ Size shown on item not same as specified on shipping container 1/

1/ For these defects, two containers shall be examined from the sample.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. Tests shall be conducted on cemented seam cut from three (3) finished hats randomly selected from each assembled shipping lot. Failure of any of the specimens tested to meet either the individual or average test requirements shall be cause for rejection of the lot. Test reports shall contain the individual test values.

Table V - Testing of cemented seams in end item

Characteristic	Requirement Paragraph	Test Method	No. of Tests Per Hat
Seam Strength	3.9	5100 1/	1
Seam Adhesion	3.9	5960 2/	2
Hydrostatic resistance:			
initial	3.9	5512 3/	1
after 3 launderings (cotton)	3.9	5556 and 5512 3/	1
after 1 dry cleaning	3.9	5508 and 5512 3/	1

1/ The seam shall be centered equidistant from the clamps and at right angles to the direction of load application.
2/ The results shall be calculated and reported to the nearest 0.1 pounds per inch by dividing the actual width of the seam into the acutal test result recorded in pounds.
3/ Water pressure shall be applied to the lightly (green) coated side with the seam centered directly over the opening.

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5. PACKAGING

5.1 Preservation - packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The topside of each hat shall be lightly dusted with the dusting powder specified in 3.3.8. The brim and neck shield of each hat shall be turned up. Eight hats of one size only shall be nested with the fronts of all hats facing the same direction. The topside of each hat shall be completely covered with a sheet of tissue paper during the nesting process. The bottom hat of each stack shall be stuffed with crumpled tissue paper.

5.1.2 Level C. Hats shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A Packing. Eight hats of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to type SF, class weather-resistant, grade V2s, size 1A of MIL-B-17757. Level A packages shall be packed upright within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

* 5.2.2 Level B Packing. Eight hats of one size only, packaged as specified in 5.1 shall be packed in fiberboard shipping container assembled and closed conforming to type CF, class domestic, variety DW, grade 200, size 1A of MIL-B-17757. Level A packages shall be packed upright within the shipping container. Each container shall have the content completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.3 Level C Packing. Hats, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed or stamped thereon, and under these words shall be legibly printed or stamped the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The hat is intended for use in conjunction with the foul weather raincoat (MIL-R-17657) and the wet-weather overalls (MIL-O-22776) for the protection of male and female naval personnel in wet or wet/cold conditions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Sizes required (see 1.2).
*c. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.
d. Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bid.

* 6.4 Water repellent treatments. Only those water repellent treatments approved by the U. S. Army Natick Research and Development Command, Natick, MA 01760 and so listed in the invitation for bids or request for proposal, shall be considered for the related procurements.

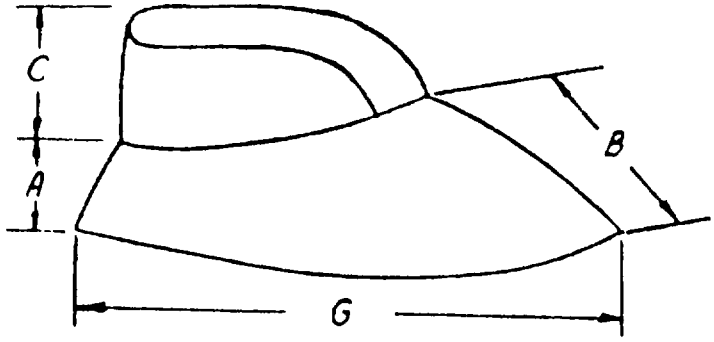
6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

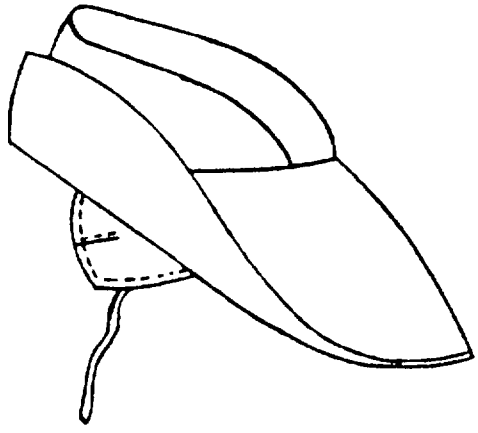
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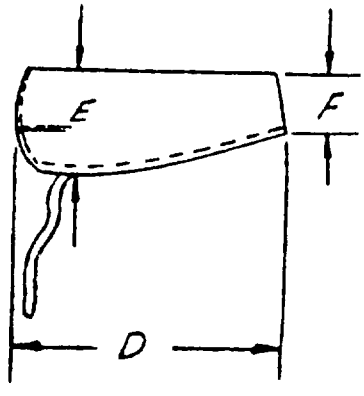


FIGURE 1 HAT, SOU'WESTER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-17656E(NU)	HAT, SOU'WESTER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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